Template 3: Data Visualizations

Visual 1: Sales by Vehicle Category

- **Chart Type:** Vertical Bar Chart
- **Target Scenario:** Scenario 1 (Emily Johnson)
- **Key Quantitative Insight:** SUVs lead with 1,255,000 units (41.83%), followed by Sedans (725,000 / 24.17%), Hatchbacks (520,000 / 17.33%), Two-Wheelers (300,000 / 10.00%), Trucks (130,000 / 4.33%), and Electric Buses (70,000 / 2.33%).

Visual 2: Top Manufacturers by Sales

- **Chart Type:** Horizontal Stacked Bar
- **Target Scenario:** Scenario 1 (Emily Johnson)
- **Key Quantitative Insight:** BYD ranks #1 globally with 785,000 units, followed by Tesla (640,000), Volkswagen (340,000), Hyundai (290,000), and Ola Electric (265,000).

Visual 3: Range vs. Satisfaction by Category

- **Chart Type:** Dual-Axis Combo Chart
- **Target Scenario:** Scenario 1 (Emily Johnson)
- **Key Quantitative Insight:** Sedans achieve the highest average range (614.7 km) and highest customer satisfaction (4.70/5), followed by SUVs (492.3 km, 4.63/5).

Visual 4: Manufacturer Market Share

- **Chart Type:** Treemap
- **Target Scenario:** Scenario 2 (David Chen)
- **Key Quantitative Insight:** Top 3 brands control 58.83% of the market: BYD (26.17%), Tesla (21.33%), and Volkswagen (11.33%).

Visual 5: Battery Tech Distribution

- **Chart Type:** Donut / Pie Chart
- **Target Scenario:** Scenario 2 (David Chen)
- **Key Quantitative Insight:** LFP dominates with 1,175,000 units (39.17%), followed by NMC (940,000 / 31.33%), NCA (480,000 / 16.00%), Sodium-Ion (255,000 / 8.50%), and Solid-State (150,000 / 5.00%).

Visual 6: Global Manufacturing Map

- **Chart Type:** Filled Geo Map
- **Target Scenario:** Scenario 2 (David Chen)

- **Key Quantitative Insight:** China leads production with 885,000 units (29.50%), followed by United States (740,000 / 24.67%), Germany (505,000 / 16.83%), and India (465,000 / 15.50%).

Visual 7: Battery Capacity vs. Range

- **Chart Type:** Side-by-Side Bar Chart
- **Target Scenario:** Scenario 3 (Dr. Sarah Williams)
- **Key Quantitative Insight:** Solid-State delivers the highest average driving range (745.0 km from 120.3 kWh), outperforming NCA (593.4 km from 193.6 kWh) and NMC (410.7 km from 140.6 kWh).

Visual 8: Customer Satisfaction Heatmap

- **Chart Type:** Highlight Table
- **Target Scenario:** Scenario 3 (Dr. Sarah Williams)
- **Key Quantitative Insight:** Solid-State Sedans (4.90/5) and Solid-State SUVs (4.80/5) earn the highest satisfaction scores, while Sodium-Ion Two-Wheelers (4.10/5) score lowest due to shorter range.

Visual 9: Country vs. Battery Tech

- **Chart Type:** Stacked Bar Chart
- **Target Scenario:** Scenario 3 (Dr. Sarah Williams)
- **Key Quantitative Insight:** China dominates LFP (640k) and Solid-State (100k) production, the United States leads NCA (480k), and Germany specializes in NMC (455k).